

Senith Jayakody

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🌐 senith-jayakody-09646b25b 🎧 SenithJayakody
📅 IESL Student Member: S-31245



Profile

Research Assistant at the Multidisciplinary AI Research Centre (MARC), University of Peradeniya, specializing in multispectral imaging systems, biomedical signal processing, and machine learning. Experienced in integrating sensing hardware, signal processing, and predictive modeling to develop robust, interpretable AI systems for environmental and healthcare applications. Strong background in experimental design, algorithm development, embedded system implementation, and academic publishing.

Research Interests

Machine learning for sensing systems, signal processing, computational imaging, biomedical time series analysis, interpretable and robust representation learning, multispectral imaging for environmental and agricultural monitoring, edge AI systems for real-world deployment.

Experience

2025 – Present

📌 **Research Assistant**, Multidisciplinary AI Research Centre (MARC), University of Peradeniya.

- Leading research on preterm birth prediction using uterine electrophysiological signals, focusing on feature extraction, model robustness, and interpretability.
- Designing and developing a dedicated biomedical data acquisition device to collect term and preterm pregnancy recordings to address dataset scarcity.
- Supervising junior researchers in multispectral imaging applications and algorithm development.
- Contributing to the development of a field deployable and commercially scalable multispectral imaging system, improving calibration, reliability, and embedded integration.

2023 – 2025

📌 **Undergraduate Researcher**, Multidisciplinary AI Research Centre (MARC), University of Peradeniya.

- Developed soil moisture estimation models using reflectance multispectral imaging.
- Extended modeling to multi soil generalization and robustness validation.
- Conducted soil texture classification and Microcystis toxicity prediction studies using MSI.
- Improved multispectral imaging hardware through lighting redesign, circuit updates, and Raspberry Pi based embedded integration.

Jul 2024 – Oct 2024

📌 **Research & Development Intern**, Swartup Pvt Ltd.

Migrated embedded systems from ESP32 FreeRTOS to NRF5340 Zephyr and designed modular hardware platforms with LoRa and GSM capability.

Experience (continued)

Jul 2023 – Oct 2023  **R&D Intern**, Sri Lanka Telecom Digital Lab.
Developed IoT based environmental monitoring systems using ESP32, MQTT, and sensor integration.

Education

2019 – 2025  **B.Sc. Eng. (Hons.)**, University of Peradeniya
Specialization: Electrical and Electronic Engineering
GPA: 3.85

2019  **G.C.E. Advanced Level**, Trinity College Kandy
Z-score: 2.16

Research Publications

Journal Articles

- 1** **S. Jayakody** et al., “Spectrotemporal feature extraction in ehg signals and tocograms for enhanced preterm birth prediction,” Manuscript submitted to IEEE Access, under review. Preprint available at arXiv:2509.07432., 2025.  URL: <https://arxiv.org/abs/2509.07432>.
- 2** G. A. S. L. Ranasinghe et al., “Reflectance multispectral imaging for soil texture composition estimation and usda texture classification,” Manuscript submitted to IEEE Access, under review., 2025.

Conference Proceedings

- 1** S. Perera, S. Ranasinghe, **S. Jayakody**, B. Epakanda, R. Godaliyadda, and M. P. Ekanayake, “Feature conditioned diffusion for audio generation,” in *IEEE 19th International Conference on Industrial and Information Systems (ICIIS)*, Accepted. To appear., 2025.
- 2** G. Thilakarathne et al., “Multispectral imaging and machine learning for soil classification through moisture-induced spectral dynamics,” in *IEEE 19th International Conference on Industrial and Information Systems (ICIIS)*, Accepted. To appear., 2025.
- 3** S. Ranasinghe et al., “Deep learning for soil moisture content estimation via reflectance multispectral imaging,” in *2024 International Conference on Advances in Technology and Computing (ICATC)*, 2024, pp. 1–6.  DOI: 10.1109/ICATC64549.2024.11025290.

Technical Skills

Programming	 Python, C/C++, MATLAB, Assembly
Machine Learning	 Scikit-learn, TensorFlow, PyTorch, Signal feature engineering, Model evaluation and validation
Signal Processing	 Time-frequency analysis, Spectral analysis, MFCC, Feature extraction, Statistical modeling
Embedded Systems	 Raspberry Pi, ESP32, NRF5340, Microcontroller firmware development
Hardware & Design	 Circuit design, Sensor integration, Multispectral imaging systems, Calibration and system validation
Tools	 Linux, Git, MQTT, Firebase, OpenCV, Altium Designer, EasyEDA

Miscellaneous Experience

Awards and Achievements

- 2025  **Second Runner-Up**, Manamperi Award 2025, Best Undergraduate Project in Sri Lanka. Final Year Project on multispectral imaging.
- 2024  **Silver Award**, Engineering Category, Sahasak Nimavum Inventions and Innovations Competition 2024. Multispectral imaging project.
-  **Top 10 Finalist**, Predicta v1.0 – IEEE Student Branch, University of Peradeniya
- 2023  **Finalist**, IEEE Electronic Design Competition – IEEE Sri Lanka Section
-  **2nd Place**, ACES Hackathon 2023 – University of Peradeniya
- 2022  **4th Place**, IESL Robogames – University Category

Certification

- 2024  **Unsupervised Learning, Recommenders, and Reinforcement Learning**
Stanford University (Coursera)
-  **Algorithmic Toolbox**
University of California, San Diego (Coursera)
-  **Introduction to Computer Vision**
Great Learning
-  **Generative AI for Beginners**
Great Learning
- 2023  **Advanced Learning Algorithms**
Stanford University (Coursera)
-  **Supervised Machine Learning: Regression and Classification**
Stanford University (Coursera)
- 2021  **Programming for Everybody (Python)**
University of Michigan (Coursera)
-  **Engineering Drawing and 3D Modelling using AutoCAD**
TecView Institute

Completed Projects

- 2024  **Soil Moisture Content Estimation via Multispectral Imaging**
Published in IEEE ICATC 2024. Developed a reflectance-based imaging system for agricultural water stress monitoring.
- 2023  **Non-Invasive Blood Glucose Sensor**
Designed a wearable prototype using optical sensors and signal processing techniques to estimate blood glucose levels without skin penetration.
-  **Survey Buddy – Sensor Fusion-based Terrain Mapping**
Created a Kalman-filter-driven system for mobile terrain mapping, integrating multiple sensors and real-time data fusion.
-  **Cyclist Safety System (GloSmart)**
An embedded safety system for cyclists designed using Assembly on Atmega328P for the EE322 Embedded Systems module.

Ongoing Projects

- 2025  **Soil Classification via Multispectral Imaging**
Using spectral signatures and neural networks to predict soil composition and type.
-  **Microcystis Cell Count and Toxicity Estimation**
Developing ML regression models for harmful algae bloom detection using MSI.

Miscellaneous Experience (continued)

■ Palm Oil Adulteration Detection in Milk

Using reflectance and transmittance MSI to detect non-dairy adulterants.

■ Preterm Birth Detection using EHG Signals

Collaborative project at the Multidisciplinary AI Research Center; focuses on term/preterm classification using MFCC features and classifiers.

■ Diffusion-Based Audio Generation Model

Developing the audio generation capabilities of DiffWave by integrating and evaluating different signal processing techniques for improved performance.

Services

2021 – 2022

■ Logistics Team Member, IEEE RAS Student Chapter, University of Peradeniya

Assisted in organizing robotics competitions and technical workshops for undergraduates.

2022 – 2023

■ Volunteer, IET on Campus, University of Peradeniya

Participated in university outreach, technical sessions, and student engagement initiatives.

Workshops & Exhibitions

2025

■ Diamond Jubilee AI Workshop for Pre-Engineering Students

Conducted sessions and interactive demonstrations on AI applications and embedded systems for A/L students.

■ AI Symposium, Trinity College Kandy

Participated in panel discussions and student sessions showcasing real-world AI use cases.

■ DreamEx 2025 – Educational Exhibition, Hatharaliyadda National School

Represented university research and demonstrated multispectral imaging applications for agriculture and health.

References

Prof. Roshan Godaliyadda

Professor

Department of Electrical and Electronic Engineering
University of Peradeniya

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Prof. Vijitha Herath

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